

Dr. Natalia Zajac

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Education

Dr. sc. ETH - ETH Zurich, Switzerland **01.09.2016 – 17.12.2020**
Department of Environmental System Science and Eawag, Swiss Federal Institute of Aquatic Science and Technology

- Title of doctoral thesis: *The genome of the trematode parasite Atriophallophorus winterbourni*, Blasco-Costa et al., 2019: A macro-and microevolutionary perspective

M.Sc. thesis - University of Bern, Switzerland **07.09.2015 – 30.05.2016**
Institute of Ecology and Evolution at, with Prof. Laurent Excoffier, funded through Swiss-European Mobility Program

- Title of Master thesis: *Chromosomal inversions in the human genome: prevalence, geographic distribution and selection pressures.*

M.Sc. in Evolutionary Biology - University of Uppsala, Sweden **01.09.2014 – 08.06.2016**
Center of Evolutionary Biology

B.A. in Biological Sciences - University of Oxford, UK **01.10.2011 - 20.06.2014**

- Title of third year dissertation: Characterization of a hybrid zone of the Common Wall Lizard, *Podarcis muralis*, in Italy
- received a Southern Field Studies Book Prize
final grade: 2.1

IB Diploma, British International School of Cracow **01.09.2009 - 30.06.2011**

Research experience

Postdoctoral Researcher **01.02.2025 – PRESENT**

Max Planck Institute for Evolutionary Biology

Mentors: Nathalie Feiner, Tobias Uller

Topic: Evolution of genome architecture in Mediterranean Wall Lizards

Bioinformatics Scientist **01.10.2021 – 31.01.2025**

Functional Genomics Center Zurich (ETH and University of Zurich)

Mentors: Dr. Hubert Rehrauer, Dr. Weihong Qi

Responsibilities:

- Bioinformatics support and data analysis in the area of Next Generation Sequencing with special focus on long read data analysis
 - advice on design of experiments using long read sequencing
 - computational and statistical analysis of next generation sequencing data
 - education and training of users (PhD students and postdocs) in software usage and data interpretation
 - Topics: Population Genomics, Genome Assemblies and Annotations, WGS, bulk RNA-seq, long read scRNA-seq, metagenomics
- Operation and management of data analysis software for long reads data analysis

- installation of software
- Processing and quality control of raw data of long read platforms
 - continuous automatic or semi-automatic processing of raw instrument data
 - consistent registration and documentation in LIMS

Postdoctoral Researcher **01.04.2021 – 30.09.2021**
Department of Computational Biology, University of Lausanne
 Supervisors: Prof. Christophe Dessimoz, Dr. Natasha Glover
 Topic: Comparative genomics of parasitic platyhelminthes with a focus on gene duplication and gene synteny

Doctoral Researcher - ETH Zurich **01.09.2016 – 30.03.2021**
Doctor of Science degree at the Department of Environmental System Science and the Institute of Aquatic Ecology, Eawag
 Supervisors: Prof. Jukka Jokela, Dr. Hanna Hartikainen
 Topic: Genomics of adaptation of *Atriphallophorus winterbourni* trematode parasite to its host, *Potamopyrgus antipodarum*

Master thesis project – University of Bern **07.09.2015 – 30.05.2016**
Institute of Ecology and Evolution
 Supervisors: Prof. Laurent Excoffier and Dr. Isabelle Dupanloup Duperret
 Topic: Characterizing the prevalence and geographic distribution of common inversions in human populations using known inversions and structural variant databases

Research assistant – University of Oxford **01.11.2013 – 10.06.2014**
Department of Zoology
 Supervisor: Dr. Tobias Uller, Sozos Michaelides
 Topic: Genetics of the common wall lizard, *Podarcis muralis*

Teaching

2025

- Guest Lecturer at the course Research Cycle in Genomics at **University of Zurich**

2021 – 2025

- **BIO253 at University of Zurich** – Research Cycle in Genomics – Fall Semester Block course for Master and Bachelor students from University of Zurich and ETH (3.5 weeks); teaching partner: Dr. Jonas Grossmann. See: <https://www.news.uzh.ch/en/articles/news/2023/future-of-teaching.html>
- **BIO638 at FGCZ** – Next Generation Sequencing Applied to Metagenomics - Spring and Fall Semester (3 day data analysis course for PhD students and Postdocs). See: https://gitlab.bfabric.org/genomics-courses/bio638_2024_june

2019

- **Ecology and Evolution**: Term Paper (701-1460-00 A)

Supervised Students

2016 – 2020

- 6 months: Co-supervision of Master Thesis of Julia Vrtilek together with Prof. Jukka Jokela
- Ecology and Evolution: Term Paper (701-1460-00 A) – supervision of Michael Zehnder
- Lab internship (3 months) - supervision of Michael Zehnder and Rebecca Oester

Prizes & Grants

Micro_innovation UZH Teaching Fund (ULF) June 2024
Best poster award (among 6 other students), Biology19

Most active and attentive audience participant, PhD symposium Eawag, 2018

Best junior talk at PhD symposium at Eawag, 2017

Swiss International Mobility Program Scholarship, 2015 – obtained from University of Bern for Master thesis project

Diploma for participating in Diana Science Conference as a Speaker, Feedback Provider and Audience, 2014

Southern Field Studies Book Prize, 2014 – obtained for the best fieldwork project of the year for third year honours research project

David Kirby Memorial Fund 2013 - fund awarded for Honours Research Project

Demyship Scholarship (Magdalen College Oxford) 2012-2013 – awarded to students obtaining a top grade (first) in end of year exams

BP-NRE award, 2012 – scholarship obtained for the Oxford University International Internship Program, awarded to support students undertaking internships in Natural Resources and the Environment

Selected Conference Talks

Talk at a Workshop on Pangenomics in Memphis	05.2025
Talk at ABRF	04.2024
<i>The impact of PCR duplication on RNA-seq data generated using NovaSeq 6000, NovaSeq X, AVITI and G4 sequencers.</i>	
Talk at the Zürich Seminars in Bioinformatics	04.2024
<i>Comparison of Single-cell Long-read and Short-read Transcriptome Sequencing of Patient-derived Organoid Cells of ccRCC: Quality Evaluation of the MAS-ISO-seq Approach.</i>	
Talk at the Swiss Institute of Bioinformatics Days	06.2020
<i>Gene duplication and gain in <i>Atriophallophorus winterbourni</i> and other major parasitic trematodes contributes to adaptation</i>	
Talk at the Biology20 Symposium in Fribourg	02.2020
<i>Gene duplication and gain in the Trematode <i>Atriophallophorus winterbourni</i> contributes to adaptation to parasitism</i>	

Courses

2021 - PRESENT

Swiss Institute of Bioinformatics Courses:

- Gene Ontology Enrichment Analysis
- Single Cell Transcriptomics
- Version Control with Git
- Docker and Singularity for reproducible research: getting started with containers
- Introduction to Snakemake for reproducible analyses
- Introduction to High-performance Computing
- Avoiding Pitfalls and Misinterpretation during the Taxonomic Profiling of Metagenomes

International Genome Graph Symposium (IGGSy'22)

ECCB/ISMB:

- Microbiome analysis with Megan6
- Exploring nf-core: a best-practice framework for Nextflow pipelines
- Computational approaches for identifying context-specific transcription factors using single-cell multi-omics datasets

Publications

Main:

- Feijen, F., Zajac, N., Vorburger, C., Blasco-Costa, I., & Jokela, J. (2022). **Phylogeography and cryptic species structure of a locally adapted parasite in New Zealand.** *Molecular Ecology*, 31(15), 4112–4126.
- Zajac, N., Zoller, S., Seppälä, K., Moi, D., Dessimoz, C., Jokela, J., Hartikainen, H., & Glover, N. (2021). **Gene duplication and gain in the trematode *Atriophallophorus winterbourni* contributes to adaptation to parasitism.** *Genome Biology and Evolution*, 13(3), evab010.
- Zajac, N. (2020) **The genome of the trematode parasite *Atriophallophorus winterbourni*, Blasco-Costa et al., 2019: A macro-and microevolutionary perspective.** ETH Zurich
- Blasco-Costa, I., Seppälä, K., Feijen, F., Zajac, N., Klappert, K., & Jokela, J. (2020). **A new species of *Atriophallophorus* Deblock & Rosé, 1964 (Trematoda: Microphallidae) described from in vitro-grown adults and metacercariae from *Potamopyrgus antipodarum* (Gray, 1843)(Mollusca: Tateidae).** *Journal of Helminthology*, 94, e108.
- Michaelides, S. N., While, G. M., Zajac, N., Aubret, F., Calsbeek, B., Sacchi, R., Zuffi, M. A. L., & Uller, T. (2016). **Loss of genetic diversity and increased embryonic mortality in non-native lizard populations.** *Molecular Ecology*, 25(17), 4113–4125.
- Michaelides, S., Cornish, N., Griffiths, R., Groombridge, J., Zajac, N., Walters, G. J., Aubret, F., While, G. M., & Uller, T. (2015). **Phylogeography and conservation genetics of the common wall lizard, *Podarcis muralis*, on islands at its northern range.** *PLoS One*, 10(2), e0117113.

- Michaelides, S. N., While, G. M., Zajac, N., & Uller, T. (2015). **Widespread primary, but geographically restricted secondary, human introductions of wall lizards, *Podarcis muralis*.** *Molecular Ecology*, 24(11), 2702–2714.
- While, G. M., Michaelides, S., Heathcote, R. J. P., MacGregor, H. E. A., Zajac, N., Beninde, J., Carazo, P., Perez i de Lanuza, G., Sacchi, R., & Zuffi, M. A. L. (2015). **Sexual selection drives asymmetric introgression in wall lizards.** *Ecology Letters*, 18(12), 1366–1375.

Under review:

- Karakulak, T.*, Zajac, N.*, Bolck, H. A., Bratus-Neuenschwander, A., Zhang, Q., Qi, W., Oltra, T. C., Rehrauer, H., von Mering, C., & Moch, H. (2024). **Heterogeneous and Novel Transcript Expression in Single Cells of Patient-Derived ccRCC Organoids.** *BioRxiv*, 2023–2024. Under review for special issue in Genome Research. – (*co-first authors)
- Zajac, N., Zhang, Q., Bratus-Neuenschwander, A., Qi, W., Bolck, H. A., Karakulak, T., Carrasco Oltra, T., Moch, H., Kahraman, A., & Rehrauer, H. (2024). **Comparison of Single-cell Long-read and Short-read Transcriptome Sequencing of Patient-derived Organoid Cells of ccRCC: Quality Evaluation of the MAS-ISO-seq Approach.** *BioRxiv*, 2023–2024. Submitted to BMC Genomics (no decision yet).
- Zajac, N., Vlachos, I., Sajibu, S., Opitz, L., Wang, S., Chittur, S., Mason, C. E., Knudtson, K. L., Ashton, J. M., & Rehrauer, H. (2023). **The impact of PCR duplication on RNA-seq data generated using NovaSeq 6000, NovaSeq X, AVITI and G4 sequencers.** *BioRxiv*, 2012–2023. Under review in Genome Biology.

In prep:

- Zajac, N., Nevers, Y., Dessimoz, C., Glover, N., **The mechanism of gene duplication is not a major driver of gene length difference in Neodermata.**
- Vrtilek, J., Zajac, N., Klapper, K., Feijen, F., Carter, G.G., Jokela, J., **Late-stage infections of New Zealand mud snails by trematode parasites reduce host movement and limit host manipulation.** In prep.
- Zajac, N., Feijen, F., Jokela, J. **Genomic footprint of divergence under high gene flow in the trematode parasite *Atriophallophorus winterbourni*.** In prep.

Co-authorship due to bioinformatic analysis in the following publications:

- Emmenegger, M., Zografou, C., Dai, Y., Hoyt, L. R., Gudneppanavar, R., Chincisan, A., Rehrauer, H., Noé, F. J., Zajac, N., Meisl, G., Schneider, M. M., Nguyen, H., Höpker, K., Knowles, T. P. J., Sospedra, M., Martin, R., Ring, A. M., Leeds, S., Eisenbarth, S. C., ... Aguzzi, A. (2024). **The Cystic Fibrosis Transmembrane Regulator Controls Tolerogenic Responses to Food Allergens in Mice and Humans.** *MedRxiv*, 2024.06.17.24309019.
- John, A., Dreifuss, D., Kang, S., Bratus-Neuenschwander, A., Zajac, N., Topolsky, I., Dondi, A., Aquino, C., Julian, T. R., & Beerenwinkel, N. (2024). **Assessing different next-generation sequencing technologies for wastewater-based epidemiology.** *MedRxiv*, 2024.05.22.24306666.
- Mengis, T., Zajac, N., Bernhard, L., Heggli, I., Herger, N., Devan, J., Marcus, R., Brunner, F., Laux, C., Farshad, M., Distler, O., & Dudli, S. (2024). **Intervertebral disc**

microbiome in Modic changes: Lack of result replication underscores the need for a consensus in low-biomass microbiome analysis. *JOR SPINE*, 7(2), e1330.

- Block, H., Migh, E., Kriston, A., Kreutzer, S., Zajac, N., Totu, T., Leary, P., Rutishauser, D., Pfammatter, S., Grossmann, J., Litchfield, C., Rupp, N., Horvath, P., Moch, H. **Morphology-driven molecular profiling of isolated single cells in heterogeneous clear cell renal cell carcinomas decodes/unravels de-differentiation of aggressive rhabdoid cells and provides rationales for immunotherapy.** In prep.

References

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